

UK National Rail Transactions

Dashboard User Guide

6 pages · 17 KPI measures · January–April 2024

Getting Started

Opening the Dashboard

1. Open Power BI Desktop (version 2.0 or later).
2. Click File → Open report → Browse to Railway_DEPI_Grad_Project_final_file.pbix.
3. If prompted about data source credentials, click Continue with current credentials.
4. The dashboard opens on Page 1 — Financial Performance by default.

Navigating Between Pages

Every page includes a Page Navigator at the bottom of the screen. Click any page button to switch instantly. The six pages are:

- Financial Performance — Revenue & pricing overview
- Tickets Details — Ticket demand & passenger behavior
- Delayed Rides — Delay analysis & root causes
- Cancelled Rides — Cancellation analysis
- Refunds & Financial Analysis — Refund rates & revenue impact
- Forecasts — 3-month projections for key KPIs

Using Slicers

Slicers are the filter panels on each page. Click any value to filter — all visuals on the page update instantly. To select multiple values, hold Ctrl and click. To clear a slicer, click the eraser icon in its top-right corner.

Important: slicers are page-specific. Selecting January on the Financial Performance page does not affect the Forecasts page.

Cross-Filtering Between Visuals

Clicking any bar, slice, or data point in a chart filters all other visuals on the same page. For example, clicking 'Weather' in the Delay Reasons bar chart on Page 3 will filter the line chart and donut chart to weather-related delays only. Click the same item again, or click empty space on the chart, to clear the cross-filter.

Resetting All Filters

To reset all slicers and cross-filters on any page: go to the View menu → Reset to default. Alternatively, clear each slicer individually using its eraser icon.

Page-by-Page Guide

Each section below describes one dashboard page: what it shows, which visuals it contains, which slicers are available, and tips for getting the most from it.

Page 1: Financial Performance

Provides a high-level view of revenue across all four months. Use this page to understand total revenue, month-over-month trends, and how revenue is split by ticket type and payment method.

Visuals on this page:

Visual	What it shows
Total Revenue card	Shows the total gross revenue for the selected period (default: all months = £741,921).
Monthly Revenue card	Revenue for the currently selected month in the Month slicer.
MoM Revenue Change (£) card	Absolute revenue difference vs. the previous month. Negative = decline.
MoM Revenue Change (%) card	Percentage change vs. the previous month. BLANK when no prior month exists.
Total Tickets card	Total ticket count for the selected filters.
Revenue by Month (line chart)	Monthly revenue trend Jan–Apr 2024. Shows the February dip and March recovery.
Revenue by Ticket Type (donut)	Revenue split: Advance (41.7%) · Off-Peak (30.1%) · Anytime (28.2%).
Revenue by Payment Method (bar)	Revenue by Contactless, Credit Card, and Debit Card.
Revenue: Weekend vs Weekday (column)	Compares revenue on weekdays vs weekends.

Slicers / filters on this page:

Slicer	Effect
Month	Filters all visuals to a specific Year-Month (e.g., Jan 2024). Select one month to see MoM cards update.
Journey Status	Filters by On Time / Delayed / Cancelled. Useful to see revenue from disrupted journeys only.

Usage tips:

- Select January in the Month slicer first — then February — to see the mom change cards update live.
- Filter Journey Status to 'Delayed' to see how much revenue came from delayed journeys (£126,814).
- The donut chart updates with all other slicers — use it to see Advance Vs Anytime revenue in different months.

Page 2: Tickets Details

Explores ticket demand patterns: which ticket types and classes are most popular, when passengers travel, how they pay, and how tickets break down by journey outcome.

Visuals on this page:

Visual	What it shows
Total Tickets card	Total ticket count under current filters (all = 31,653).
Avg Price card	Average ticket price under current filters (all = £23.44).
Ticket Class Distribution (pie)	Standard (90.3%) vs First Class (9.7%) split.
Ticket Sales by Purchase Type (donut)	Online vs. Station ticket purchases.
Ticket Count & Avg Price by Type (combo)	Column chart (count) + line chart (avg price) by ticket type. Anytime has the highest avg price.
Tickets by Departure Time Segment (ribbon)	Demand across Morning / Afternoon / Evening / Night.
Ticket Count by Payment Method (bar)	Number of tickets purchased by each payment method.
Avg Price by Ticket Type (column)	Average price comparison: Advance vs Off-Peak vs Anytime.
Decomposition Tree	Drill from Total Tickets → Journey Status → Reason for Delay → Refund Request. Expand each node to drill deeper.

Slicers / filters on this page:

Slicer	Effect
Month	Filter to a specific month to see demand patterns in that period.
Purchase Type	Filter to Online or Station purchases.
Journey Status	Filter to On Time / Delayed / Cancelled.
Payment Method	Filter to Contactless, Credit Card, or Debit Card.
Ticket Type	Filter to Advance, Off-Peak, or Anytime.

Usage tips:

- Use the Decomposition Tree to answer questions like: 'Of all Advance ticket holders, how many were delayed and then requested a refund?'
- Filter Payment Method to Debit Card — notice it represents the smallest revenue segment (£52,966).
- The Ribbon Chart shows which departure time segments are most popular in each selected month.

Page 3: Delayed Rides

Analyses journey delays in detail: how many journeys were delayed, which reasons are most common, which stations are worst affected, and whether delay patterns differ by ticket type or class.

Visuals on this page:

Visual	What it shows
Delayed Rides card	Count of journeys with status 'Delayed' (all = 2,292).
On-Time Rides card	Count of on-time journeys (all =29,380).
Cancelled Rides card	Count of cancellations (all = 1,880) — shown here for context.
Off Time Rides % card	Percentage of all journeys that were delayed or cancelled (13.2%).
Sum Actual Delay (Min) card	Total accumulated minutes of delay across all delayed journeys.
Delayed Rides over time (line)	Monthly trend of delay counts.
Delayed Rides by Payment (donut)	Payment method split among delayed journeys.
Delayed Rides by Reason (bar)	Top causes of delay: Weather, Technical Issue, Signal Failure, Staffing.
Delayed Rides by Ticket Class (column)	Standard vs First Class breakdown of delayed journeys.
Delayed Rides by Ticket Type (column)	Advance vs Off-Peak vs Anytime breakdown.

Slicers / filters on this page:

Slicer	Effect
Journey Status	Pre-set to 'Delayed'. Keep this selection to see delay-specific data.
Stations (Departure)	Filter to a specific departure station to see its delay profile.
Arrival Station	Filter to a specific arrival destination.
Reason for Delay	Filter to a specific delay cause (e.g. Weather only).

Usage tips:

- Filter Departure Station to 'Manchester Piccadilly' (the busiest hub) to see how it compares to the network average.
- Select 'Weather' in the Reason for Delay slicer to isolate weather-related delays — these are the largest group.
- Combine Departure Station + Reason for Delay to identify the most problematic route-reason combinations.

Page 4: Cancelled Rides

Mirrors the Delayed Rides page in structure but focuses on cancellations. Use this to identify which routes, reasons, and ticket types are most affected by full cancellations.

Visuals on this page:

Visual	What it shows
Cancelled Rides card	Count of cancelled journeys (all = 1,880).
On-Time Rides card	On-time count — shown for comparison.
Delayed Rides card	Delay count — shown for comparison.
Off Time Rides % card	Combined disruption rate.
Sum Actual Delay (Min) card	Delay minutes (relevant for partially-cancelled journeys).
Cancelled Rides over time (line)	Monthly trend of cancellation counts.
Cancelled Rides by Payment (donut)	Payment method split among cancelled journeys.
Cancelled Rides by Reason (bar)	Reasons behind cancellations.
Cancelled Rides by Class (column)	Standard vs First Class cancellation split.
Cancelled Rides by Type (column)	Advance vs Off-Peak vs Anytime cancellation split.

Slicers / filters on this page:

Slicer	Effect
Journey Status	Pre-set to 'Cancelled'. Keep this selection.
Stations (Departure)	Filter by departure station.
Arrival Station	Filter by arrival destination.
Reason for Delay	Filter by cancellation reason.

Usage tips:

- Compare the Reason for Delay bar chart on this page vs Page 3 — some reasons cause delays; others cause outright cancellations.
- Filter by Ticket Type = 'Anytime' to see which Anytime ticket holders were most impacted by cancellations.
- Use both pages side-by-side by opening two browser tabs to compare delay vs cancellation profiles.

Page 5: Refunds & Financial Analysis

Analyses the financial impact of disruptions through refund requests. Shows total refunded revenue, refund request rates, monthly trends, and which stations generated the most refund claims.

Visuals on this page:

Visual	What it shows
Total Revenue Lost card (Refunded Revenue)	Total value of tickets where a refund was requested (£38,702).
Total Refunds card	Number of refund requests submitted (1,118).
Revenue After Refunds card	Net revenue after deducting all refunded values (£703,219).
Refund Rate by Reason (line)	How refund request rate varies by delay reason.
Refund Request Number by Month (line)	Monthly trend of refund request counts.
Refunded Revenue (Delay) by Station (column)	Revenue refunded from delayed journeys, by departure station.
Refunded Revenue (Cancelled) by Station (column)	Revenue refunded from cancelled journeys, by departure station.

Slicers / filters on this page:

Slicer	Effect
Month / Months	Filter to specific months to see refund trends in that period.
Journey Status	Filter to Delayed or Cancelled to compare refund patterns by disruption type.
Reason for Delay	Filter to a delay reason to see how refund behavior changes (e.g., Signal Failure vs Weather).

Usage tips:

- Filter Journey Status to 'Cancelled' — note the refund rate is higher for cancellations than delays, as passengers have stronger entitlements.
- Compare the two station column charts to identify which stations generate the most refund pressure.
- The refund rate by reason line chart reveals whether certain causes (e.g., Signal Failure) prompt more claims than others (e.g., Weather).

Page 6: Forecasts

Uses Power BI's built-in forecasting analytics to project revenue, ticket demand, delayed rides, and cancelled rides for the next three months beyond the dataset (May–July 2024). Confidence intervals are shown on each line chart.

Visuals on this page:

Visual	What it shows
Revenue Forecast (line)	Projects monthly revenue through Jul 2024 based on Jan–Apr trend. Shaded area = confidence interval.
Ticket Demand Forecast (line)	Projects monthly ticket volumes through Jul 2024.
Delayed Rides Forecast (line)	Projects the number of delayed journeys per month.
Cancelled Rides Forecast (line)	Projects the number of cancelled journeys per month.

Slicers / filters on this page:

Slicer	Effect
Month	Filter the base data range used to generate the forecast.
Journey Status	Filter to see forecasts for specific journey outcome subsets.
Day Name	Filter to a specific day of the week to see its individual demand forecast.
Weekend vs Weekday	Toggle between weekday-only and weekend-only forecasts to compare seasonal patterns.

Usage tips:

- Filter Weekend vs Weekday to 'TRUE' (weekend) to see whether weekend demand is projected to grow faster or slower than weekday.
- Select a specific Day Name (e.g., Friday) to see which day has the highest projected ticket volumes.
- The confidence interval bands widen as the forecast extends further — this reflects increasing uncertainty. Do not treat the outer bounds as targets.
- These forecasts are indicative only; they are based on 4 months of data and do not account for seasonal events, bank holidays, or network changes.

KPI Reference

The table below lists every KPI card and measure displayed in the dashboard, along with its default value (no filters applied) and a plain-English definition.

KPI Name	Default Value	Definition
Total Revenue	£741,921	SUM of all ticket prices — the headline revenue figure.
Monthly Revenue	Varies	Revenue scoped to the month selected in the Month slicer.

KPI Name	Default Value	Definition
MoM Revenue Change (£)	Varies	Absolute revenue vs. prior month. Negative = revenue declined.
MoM Revenue Change (%)	Varies	Percentage change vs. prior month. BLANK if no prior month data.
Total Tickets	31,653	Count of all ticket transactions in the selected filter context.
AVG Price	£23.44	Mean ticket price across all transactions.
On-Time Rides	29,380	Journeys arriving on schedule (Journey Status = On Time).
Delayed Rides	2,292	Journeys where actual arrival exceeded scheduled arrival.
Cancelled Rides	1,880	Journeys that did not operate.
Off Time Rides %	13.2%	Percentage of all journeys that were disrupted (delayed + cancelled).
Sum Actual Delay (Min)	Total	Accumulated minutes of delay across all delayed journeys.
Refund Request Number	1,118	Count of tickets where a refund was requested.
Refunded Revenue	£38,702	Total ticket value of all refund-requested journeys.
Revenue After Refund	£703,219	Net revenue = Total Revenue minus Refunded Revenue.
Refunded Revenue (Delay)	Varies	Refunded revenue from delayed journeys, filterable by station.
Refunded Revenue (Cancelled)	Varies	Refunded revenue from cancelled journeys, filterable by station.

Troubleshooting

Issue	Resolution
MoM cards show BLANK or 0	The Month slicer is set to January (no prior month). Select February, March, or April, or clear the slicer to see relative changes.
Forecast lines are missing	The Forecasts page requires a continuous date axis. Ensure no day-level slicer is filtering out intermediate dates. Clear all slicers and try again.
Visuals show 'No data'	A slicer combination has been applied that returns no records. Clear one or more slicers (click the eraser icon) to restore data.
Decomposition Tree not expanding	Click the '+' icon on any node to expand it. The tree drills from Total Tickets → Journey Status → Reason for Delay → Refund Request.
Dashboard runs slowly	Close other open Power BI files or browser tabs. The dataset has 31,653 rows which should load in under 5 seconds on standard hardware.

Quick Reference

Keyboard Shortcuts in Power BI Desktop

Shortcut	Action
Ctrl + Z	Undo last action
Ctrl + Shift + F	Spotlight / focus on a visual
Ctrl + click slicer	Select multiple slicer values at once
Escape	Clear a cross-filter or exit focus mode
Tab	Navigate between visuals (accessibility)

Dataset Summary

Property	Value
File name	railway.csv
Total rows	31,653 ticket transactions
Columns	18
Date range	January 2024 – April 2024 (journey dates)
Departure stations	Manchester Piccadilly, London Euston, Liverpool Lime Street, London Paddington, London Kings Cross and others
Total revenue	£741,921
Avg ticket price	£23.44
On-time rate	86.8% (27,481 journeys)
Disruption rate	13.2% (2,292 delayed + 1,880 cancelled)